



Vaibhav Nawal
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B.Sc. Electrical
Gender: Male
DOB: 03-06-2000



Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	7.69

Analytical, user-centric **IIT Bombay** graduate with strong **operational** and client-facing mindset. Proven in **structured** problem solving, **data-driven** decision making, and **cross functional** stakeholder coordination. Seeking *Strategy, Operations or Managerial* roles to drive growth, improve operational experience, and deliver **measurable** business impact.

ACADEMIC ACCOLADES	
JEE	- JEE-Adv. '18- top 3 % among 0.3 million students - JEE Mains '18 - 99.4+ %ile among 1.3 million+
UNDER GRADUATION	- Achieved grades of 9.0 or higher in over 50% of completed credits - Highest grade AA (10/10) grade in a rigorous analytical materials modeling course
KEY PROJECTS	
SUPERVISED RESEARCH EXPOSITION	<u>Development of an Ear and Fingerprint-Based Biometric Recognition Pipeline</u> [July '23 – Nov '23] - Designed a classification system that improved differentiation accuracy by up to 40%, enabling more reliable identification decisions - Achieved accuracy of 98% across a test set of 300 queries, as visualized using Cumulative Match Curve - Analysed 150,000+ data points to define decision thresholds, balancing accuracy against risk of false positives - Improved data quality across 800+ inputs by standardizing preprocessing steps, reducing noise by 25% - Designed and evaluated multiple decision frameworks to minimise false positives below 5%, demonstrating risk management and quality control discipline
DEEP LEARNING	<u>Exploring Multi-Modal Deep Learning: Enhancing Vision Based DML and OOD Detection</u> [June '22 – July '23] - Introduced novel Hint Learning for DML, achieving loss reduction across 80+ training epochs using <i>intermediate</i> layer features - Achieved 2% improvement in few-shot accuracy across 11+ benchmarks by extending Maximum Concept Matching with Prompt Learning - Achieved 99.62% AUROC and 1.70% FPR95 on the Oxford-Pet dataset engineering multi-modal OOD detection - Trained 3x faster, reaching peak performance in 50 epochs instead of 150, and open-sourced all 100+ experiments on GitHub. - Raised overall accuracy (F1) by 18% by guiding the network with intermediate “hints” from a language model - Cut false-alarm rate from 35% to 4% by scoring examples with combined vision-language signals
	- Presented <i>complex</i> insights to non-technical stakeholders, influencing final decisions - Structured recommendations that <u>improved</u> adoption of proposed solutions - Demonstrated <i>persuasive</i> communication backed by data and clarity
EXTRACURRICULAR ACTIVITIES	
BOOTCAMPS & COURSES	- Financial Markets – Yale University (Robert Shiller) - Marketing & Finance for Entrepreneurs – DSSE IIT Bombay - Excel skills for business by Macquarie University
SOFTWARES	Excel PowerPoint Tableau Python MATLAB GNU Radio LaTeX
CLIENT SERVICE & BUSINESS CAPABILITIES	- Customer requirement analysis and structured problem resolution - Stakeholder <i>communication</i> and expectation management - Process improvement through data analysis and standardization Cross-functional coordination and follow-through
EXTRA CURRICULARS	- Completed Devils Circuit, (5km 15+ obstacle) known as India's toughest obstacle course - Led rapid design and execution of a functional RC car prototype under a 4-day deadline. - Designed a remote-controlled airplane from ground up with Aero Modelling club
INTERESTS	Football Chess Formula 1 Geography Tennis Weightlifting